CEBU INSTITUTE OF TECHNOLOGY - UNIVERSITY
College of Computer Studies, IT411 Capstone & Research 2

Final Validation Results and Statistical Analysis Report (`04_Results_and_Analysis`)

Week 12 Deliverable 04: findings per research question (RQ1-RQ5) and SMART goal

Course & Section:	IT411 (Section G7) — Capstone & Research 2 | Semester 1, AY 2026–2027
Group Code / Team:	`2627-sem1-it411-56` — Group 56 (`Riva, Zendrix`; `Palis, John James`; `Miel, Kaysean`; `Durano, Ashlee Skye`; `Bien, Erik Samuel`)
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Deliverable Reference:	Week 12 Deliverables 3, 4 & 5 (`03_Testing_Evidence`, `04_Results_and_Analysis`, `05_Recommendations_and_Improvements`, Due October 31, 2026)

Folder 04 Template:
`04_Results_and_Analysis/FINAL_VALIDATION_RESULTS_AND_ANALYSIS.md`

1. Respondent Profile & Sample Distribution

Cohort	Target Role	Actual N	Demographic / Qualification Summary
Cohort 1 (`C-01..C-__`)	Grade 1 Learners (Primary End-Users)	`[__]`	Ages 6–7 (`[__]` age 6, `[__]` age 7); Grade 1 emergent readers.
Cohort 2 (`T-01..T-__`)	DepEd Grade 1 Reading Teachers	`[__]`	Mean teaching experience: `[__]` years in Grade 1 Reading / MTB-MLE / English.
Cohort 3 (`P-01..P-__`)	Parents / Primary Caregivers	`[__]`	Parents/guardians of Grade 1 learners supervising home reading practice.
Cohort 4 (`QA-01..QA-03`)	Technical / QA Evaluators	3	Hardware & accessibility benchmark evaluators on `SM-A217F`.

2. Empirical Results by Research Question & SMART Goal

2.1 RQ1 — Pedagogical & Acoustic Integrity (SMART-01, SMART-02, HI-1, HI-2)

- **Gate 3 Blind Phoneme Audit (`Instrument A` Part A1):** `[__] / 26` (`[____]`%) phoneme clips transcribed accurately and rated schwa-free (`>= 2/3` teachers).
- **Decodable CVC Word Bank Audit (`Instrument A` Part A2):** `[__] / 33` (`[____]`%) Blend It words verified as 100% cumulatively decodable.
- **Child `Hear It` Independent Completion (`Instrument B` Part B1):** `[__] / [__]` sessions (`[____]`%, 95% Wilson CI `[____%, ____%]`) completed `Hear It` with `0` adult prompts *(Target: ≥ 90.0%)*.

2.2 RQ2 — Offline Speech Recognition & Articulatory Scaffolding (SMART-03, SI-1–SI-4)

Metric	Formula	Empirical Result	95% Wilson CI / Spread	Target Criterion	Verdict
Teacher-ASR Agreement (P_(agree))	(TP + TN) / N_(trials)	`[____]%` (`[__]/[__]`)	`[____%, ____%]`	**≥ 80.0%**	`[PASS / FAIL]`
False-Reject Rate (P_(FR))	FN / (TP + FN)	`[____]%` (`[__]/[__]`)	`[____%, ____%]`	**≤ 15.0%**	`[PASS / FAIL]`
Dual-Teacher Inter-Rater kappa	Cohen's kappa	`[0.__]`	—	**≥ 0.70**	`[PASS / FAIL]`
3-Attempt Lesson Cleared Rate (`SI-1`)	Cleared ≤ 3 att. / Total	`[____]%`	`[____%, ____%]`	**≥ 80.0%**	`[PASS / FAIL]`
Attempt 2/3 Mouth-Cue Recovery (`SI-2`)	Att. 2–3 pass mid Att. 1 miss	`[____]%`	`[____%, ____%]`	Documented > 0%	`[PASS / FAIL]`
`Say It` **P90 Latency (`D-01`)**	P90(t_(feedback) − t_(EOS))	`[____] ms`	`P50: [____] ms`	**≤ 500 ms**	`[PASS / FAIL]`

2.3 RQ3 — Child Usability, Discrimination & Blending Mastery (SMART-04, SMART-05, 03)

- **Zero-Prompt Child Screen Navigation (03):** `[____]`% (`[__]/[__]` transitions) completed without adult intervention *(Target: ≥ 85.0%)*.

- `Find It` **First-Tap Accuracy & Latency (**`SMART-04`**,** `D-02`**):** First-tap accuracy `[____]`%; P90 tap latency `[____] ms` *(Target: ≤ 300 ms).*
- `Blend It` **CVC Word Completion (**`SMART-04`**):** `[____]`% of learners solved all 3 group CVC words with ≥ 2star mastery.
- **Child Affect & Voluntary Replay (**`Instrument E` **Part E1):** `[____]`% selected Happy Smile (`3/3`); `[____]`% voluntarily chose to play another letter when offered the exit prompt.

2.4 `RQ4` — Hardware Performance, Offline Persistence & Privacy (`SMART-06, 04, 05`)

- **20-Cycle Force-Close Persistence (**`D-05`**):** `[20 / 20]` (`100%`) cycles retained exact SQLite state in Airplane Mode.
- **Airplane-Mode Module Operability (**`D-06`**):** `[4 / 4]` learning modules + offline PDF export (`100%`) operated without network connectivity.
- **Frame Stability & Memory (**`D-04`**):** Cold start `[____] ms` (≤ 2500 ms); Jank `[____]`% (≤ 10%); Peak PSS RAM `[____] MB` (≤ 200 MB).

2.5 `RQ5` — Stakeholder Usability (SUS), TAM & Diagnostic Utility (`SMART-07, 06`)

Stakeholder Scale / Dimension	Teachers (n = [__]) Mean (SD)	Parents (n = [__]) Mean (SD)	Combined (N = [__]) Mean (SD)	Target Criterion	Verdict
Brooke System Usability Scale (SUS, 0–100)	`[___._] ([_._])`	`[___._] ([_._])`	`[___._] ([_._])`	**≥ 75.0**	`[PASS / FAIL]`
TAM Perceived Usefulness (`PU1`**–**`PU4`**, 1–5)**	`[_. __] ([_.__])`	`[_. __] ([_.__])`	`[_. __] ([_.__])`	**≥ 4.00 / 5.00**	`[PASS / FAIL]`
TAM Perceived Ease of Use (`PEOU1`**–**`PEOU3`**, 1–5)**	`[_. __] ([_.__])`	`[_. __] ([_.__])`	`[_. __] ([_.__])`	**≥ 4.00 / 5.00**	`[PASS / FAIL]`
Pedagogical Alignment (`PA1`**–**`PA5`**, 1–5)**	`[_. __] ([_.__])`	N/A	`[_. __] ([_.__])`	**≥ 4.00 / 5.00**	`[PASS / FAIL]`
Adult Diagnostic & PDF Task Success (`Tasks A1–A4`**)**	`100%`	`100%`	`100%`	**100%**	`[PASS / FAIL]`

3. Comparison: Initial MVP Validation (Weeks 1–2) vs. Final System Validation (Week 12)

Evaluation Dimension	Initial MVP Baseline (Weeks 1–2, N = 18)	Final Refactored System (Week 12, `v1.2.0-rc2`)	Net Improvement & Engineering Root Cause
Phoneme Audio Purity (Gate 3)	`3.10 / 5.00` (Schwa /■/ on stop consonants)	`[____]`% **(26/26 Pass)**	Audited human/Kokoro/Chatterbox/ElevenLabs clips (`Cards 09a–09c, 15b, 19`).
`Say It` **ASR Child Agreement**	`2.85 / 5.00` (High false-rejects & foil acceptance)	`[____]`% **Agreement /** `[____]`% **FRR**	Constrained `SpeechValidator` grammar + foil rejection + non-punitive 3-attempt scaffolding (`Cards 02, 05, 13, 21`).
Word Blending (`Blend It`**)**	Absent in MVP (`0` words)	**33 Decodable CVC/VVC Words across 6 Groups**	Full `BlendItScreen` + non-solved tile deal guard (`Cards 08, 15, 25, 31f`).
Child Touch & Stage Ergonomics	`3.20 / 5.00` (Sub-48dp buttons, competing gradients)	**100% ≥ 64 dp targets + Duolingo ABC Grounded Stages**	`LessonScaffold`, 64dp child controls, Argon 4 illustrations, and Card 32 grounded horizon stages.